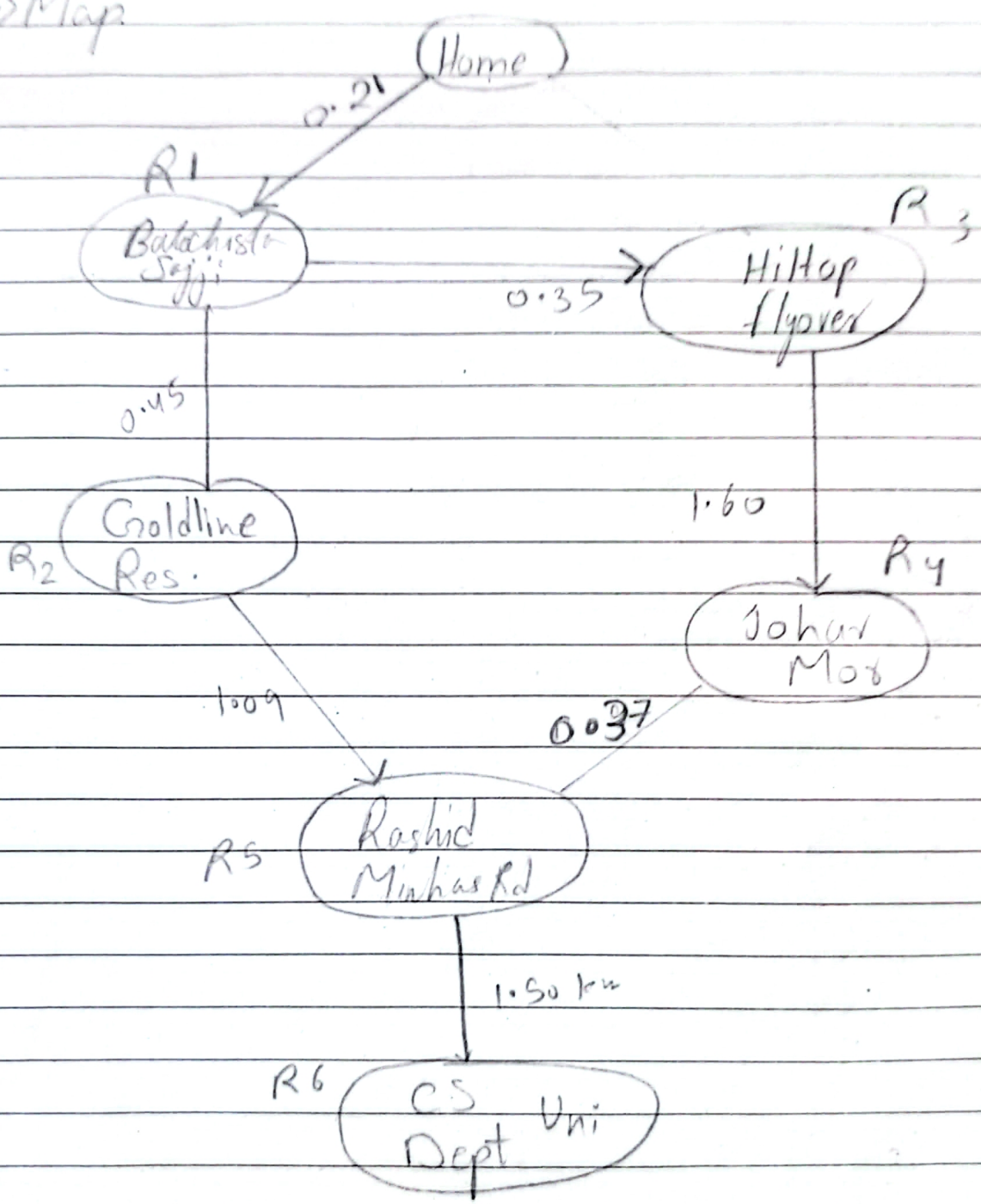
-> M. Abdullah
231222206

A* - Home - Uni

-> Map



Date: _____

M	T	W	T	F	S	S

→ Graph Structure:

Node	:	Connects to
Home	→	R_1
R_1	→	Home, R_2, R_3
R_2	→	R_1, R_5
R_3	→	R_1, R_4
R_4	→	R_3, R_5
R_5	→	R_2, R_4, R_6
R_6	→	R_5

→ Route Distance

	$h(n)$
home	2.6
R_1	2.4
R_2	2.0
R_3	2.5
R_4	1.2
R_5	1.4
R_6	0

∴ $f(n) = g(n) + h(n)$

CamScanner

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→ A* Trace:

S0 : Initialize

PQ = [Home]
V = ∅
g(Home) = 0
f(Home) = 0 + 2.6 = 2.6

(P.T)

Home → None

S1 : Expand Home.
Neighbor : R1
PQ : [R1]
V : [Home]

Node	g	h	f
R_1	1	2.6	3.6

Choose R1

PT: Home → None
R1 → Home

S2 : Expand R1
Neighbor : R_2, R_3, Home(V)
PQ : [R_2, R_3]
V : [Home, R_1]

Node	g	h	f
R_2	2	2.0	4.0
R_3	2	2.5	4.5

choose R_2

PT: Home → None
R1 → Home
R_2 → R1

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Date:

M T W T F S S

S3: Expand R3

Neighbor: [R2(V), R5]

Q: [R5, R3] V: [Home, R1, R2]

Node	g	h	f
R5	3	1.4	4.4

Choose R5

PT= Home → none

R1 → Home

R2 → R1

R5 → R2

S4: Expand R5

Neighbor: [R2(V), R6, R4]

Q: [R6, R4] R3] V: [Home, R1, R2, R5, R6]

Node	g	h	f
R6	4	0	~~5.2~~ 4
R4	4	1.2	5.2

Choose R6

PT: Home → None

R1 → Home

R2 → R1

R5 → R2

R6 → R5

S5: Expand R6

Neighbor: [R5(V)]

Q = [R3, R4]

CamScanner

Date:

M	T	W	T	F	S	S

V = [Home, R_1, R_2, R_3, R_6]

Goal found

Shortester Path:

Home → R_1 → R_2 → R_5 → R_6 (FUUAST)

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